

Mohit Kumar Sharma

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Education

Indian Institute of Technology (BHU) Varanasi
B.Tech - Electronics Engineering

2020 - 2024
CGPA - 8.42

Experience

Software Engineer

Hyderabad, India

Microsoft

July 2024 – Present

- Designed and implemented **multi-tier caching** (in-memory + persistent on-disk, asynchronous background revalidation) and parallelized codec DLL probing, reducing encoder/decoder capability-enumeration latency in a real-time communications media stack from **1–3s to 5ms** during application startup.
- Improved Microsoft Teams H.264 encoder and decoder hardware acceleration through large-scale telemetry analysis, root-cause debugging, experimentation, and production rollouts, delivering **4% improvement** in overall **decoder** hardware utilization, and **5% improvement** in overall **encoder** hardware utilization, and substantial gains in reliability while reducing software fallback rates.
- Designed and built a scalable distributed test execution and analytics system with real-time test tracking, automated failure analysis, intelligent alerting, and centralized dashboards, significantly reducing manual debugging effort and improving large-scale issue detection and overall test infrastructure reliability.
- Introduced new telemetry metrics tracking hardware failures and encoder/decoder frame drops and built **automated anomaly detection** on HW utilization and reliability KPIs, enabling early regression detection and cutting time-to-detect from days to hours.
- Developed an **AI-powered** investigation platform that automates KPI regression analysis by correlating telemetry signals with diagnostic logs, orchestrating DDL experiments, and generating root-cause insights, significantly reducing manual investigation effort and accelerating issue resolution.
- Exposure:** C++, Python, Multithreading & Concurrency, System Design, MCP, Agentic AI, KQL, Power BI

Software Engineer Intern

Hyderabad, India

Microsoft

May 2023 – July 2023

- Designed and developed multi-turn conversational interactions for the TAC Co-pilot, enhancing UI/UX by implementing intuitive chiclet-based input options and aligning the interface with the M365 Co-pilot design system.
- Leveraged the Microsoft **LLM framework** to accurately interpret user queries by identifying intents and extracting entities, improving overall conversational performance and user experience.
- Exposure:** React, TypeScript, REST APIs, UI/UX, Prompt Engineering

Full-Stack Developer Intern

Remote

Warelogg

Dec 2022 – Dec 2022

- Enhanced the Warelogg React web application by improving UI/UX, responsiveness, and overall functionality to deliver a smoother user experience.
- Implemented Node.js-based backend updates to manage and modify warehouse details, ensuring secure and accurate data handling through admin-level access controls.

Projects

Picture Manager Application [🔗](#)

- Developed a web-based application featuring a secure login system powered by Face Recognition technology, integrating **JWT-based authentication** and authorization to safeguard user/admin access and protect all API endpoints.
- Developed intelligent photo-management functionality using **facial recognition** to automatically categorize images into individual and group folders, and implemented event-related CRUD operations with MongoDB for optimized data handling.
- Exposure:** Next.js, Node.js (Express.js), Face-api.js, MongoDB, Tailwind CSS

Technologies

Languages: C/C++, Python, JavaScript, React, NodeJS, SQL, NoSQL

Core Skills: Full-Stack Development, Data Structures and Algorithms, Problem-Solving, OOPS, DBMS, System Design

Achievements

- Solved **800+ coding challenges** on platforms like LeetCode, GeeksforGeeks, showcasing strong problem-solving skills.
- Finalist in national-level technical competitions “Pixelate” (Technex’22) and “Mosaic” (Udyam’22), both organized by IIT (BHU) Varanasi.